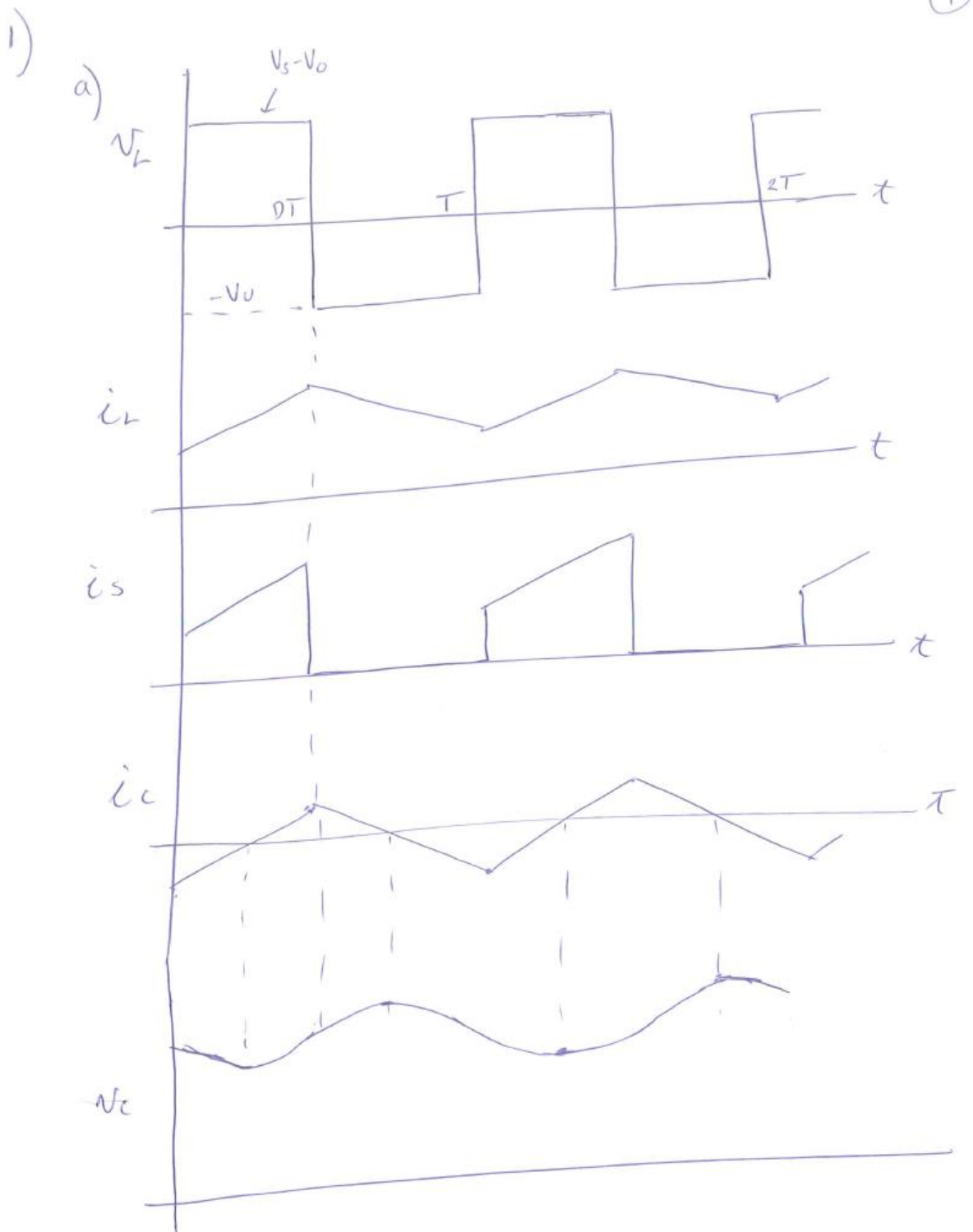


# Buck Converter Problems - Solutions

1.



1b) Rule : average inductor voltage = 0

$$\therefore (V_s - V_o)DT - V_o(1-D)T = 0$$

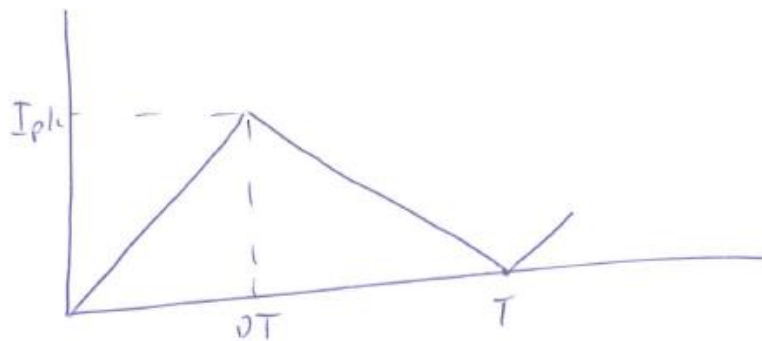
$$\therefore (V_s - V_o)D - V_o(1-D) = 0$$

$$\therefore DV_s - \cancel{V_o D} - V_o + \cancel{V_o D} = 0$$

$$\therefore V_o = D V_s$$

c) Minimum inductance for continuous conduction,  
load current = 0.6 A

Waveform, inductor current



$$\text{average value } I_o = \frac{I_{pk} T}{2T} = I_{pk}/2 \quad (1)$$

$$\text{from switch off, } I_{pk} = \frac{V_o \cdot (1-D)T}{L} \quad (2)$$

Substitute (2) into (1)

1c) cont'd

$$I_0 = \frac{V_o(1-D)T}{2L}$$

$$\text{or } L = \frac{V_o(1-D)}{2I_0 f_s}$$

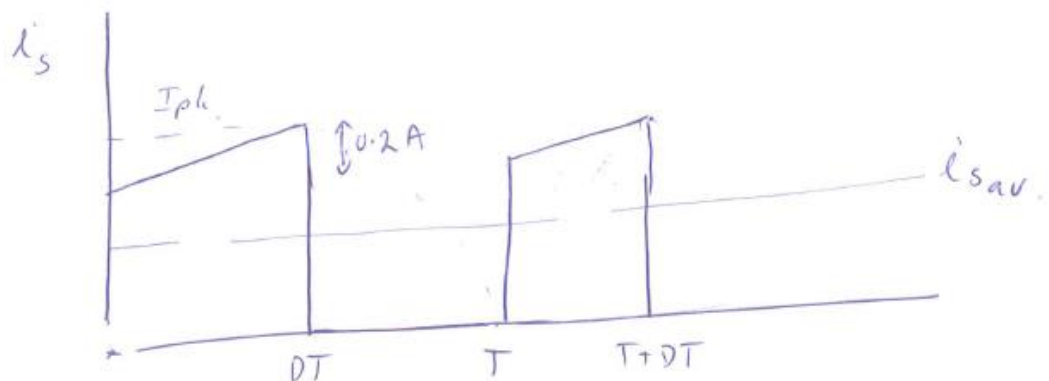
for this circuit,  $V_o = 5$ ,  $D = \frac{V_o}{V_s} = \frac{5}{12} = 0.4167$

$$I_0 = 0.6 \quad f_s = 10^5$$

$$\therefore L_{\min} = \frac{5(1-0.4167)}{2 \times 0.6 \times 10^5} = 2.43 \times 10^{-5}$$

$$\text{or } 24.3 \mu\text{H}$$

1d) draw waveform. Input current  $i_s$



The capacitor charges during the period  $(1-D)T$ .

During this time the average value of  $i_s$  is

1d) continued

by conservation of power

$$V_o I_o = V_s I_s \quad (\text{average values})$$

$$\therefore I_{s_{av}} = \frac{V_o}{V_s} I_o = D I_o = 0.4167 \times 0.6 \text{ A} \\ = 0.25 \text{ A}$$

$\therefore$  Capacitor discharges  $i = C \frac{dv}{dt} = C \frac{dV}{DT}$

$$\therefore C = \frac{i DT}{dV}$$

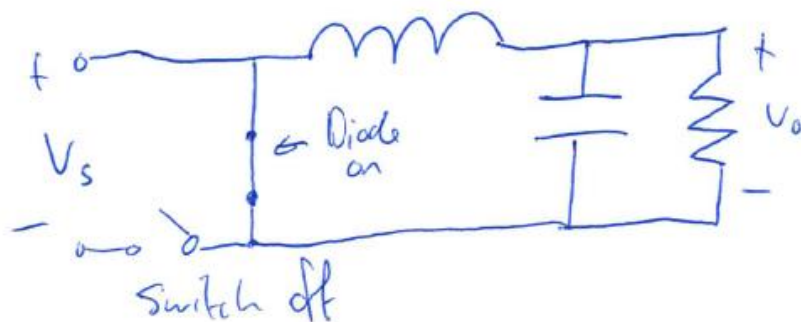
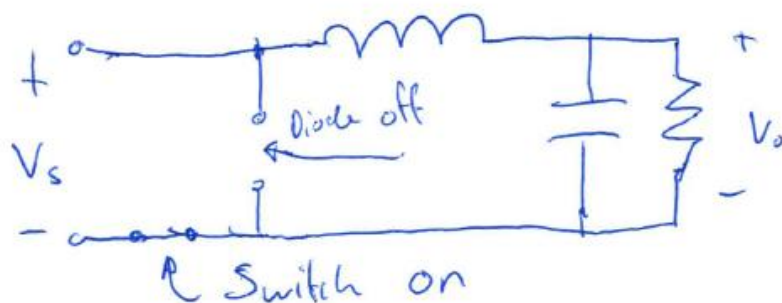
$$= \frac{I_{s_{av}} \cdot (1-D) T}{dV}$$

$$T = \frac{1}{f_s} = \frac{1}{10^5}$$

$$\therefore C = \frac{0.25 (1-0.4167)}{10^5 \times 0.5} = 2.9 \times 10^{-6}$$

$$= 2.9 \mu\text{F}$$

2)



The switch is often positioned on the low-side as this allows for easy gate driving of a MOSFET switch. This is because the MOSFET source terminal is permanently connected to the input voltage source negative terminal, and as such the required positive gate-to-source voltage of the MOSFET is easily produced with circuitry directly connected to the input voltage source.

3) At high switching frequencies, the capacitive impedance of the MOSFET gate will need to be charged and discharged very quickly in order to switch on and off quickly. As such, a dedicated gate drive circuit is needed that can source and sink pulse high current for high frequency switching applications.